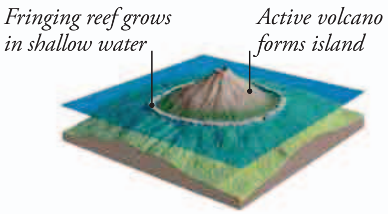
Atolls and lagoons

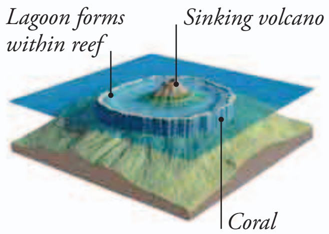
Some tropical seas are dotted with islands surrounded by coral reefs. Many of the islands are extinct volcanoes that are sinking below the waves. As they sink, the coral keeps growing, and over time, the original islands disappear to leave ring-shaped reefs called atolls, enclosing shallow lagoons. Other atolls have formed on ridges of rock submerged by rising sea levels.

SINKING VOLCANOES

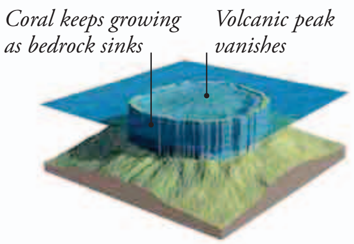
When an island volcano stops erupting, the rock beneath it cools and shrinks, so the island starts sinking. The coral around it grows upward to compensate, so the fringing reef becomes a barrier reef and eventually a ring of coral—an atoll.



▲ 1. FRINGING REEF
A tropical volcanic island soon develops a fringe of living coral near the shore.



▲ 2. BARRIER REEF
When the extinct volcano starts sinking, the coral grows up toward the light.

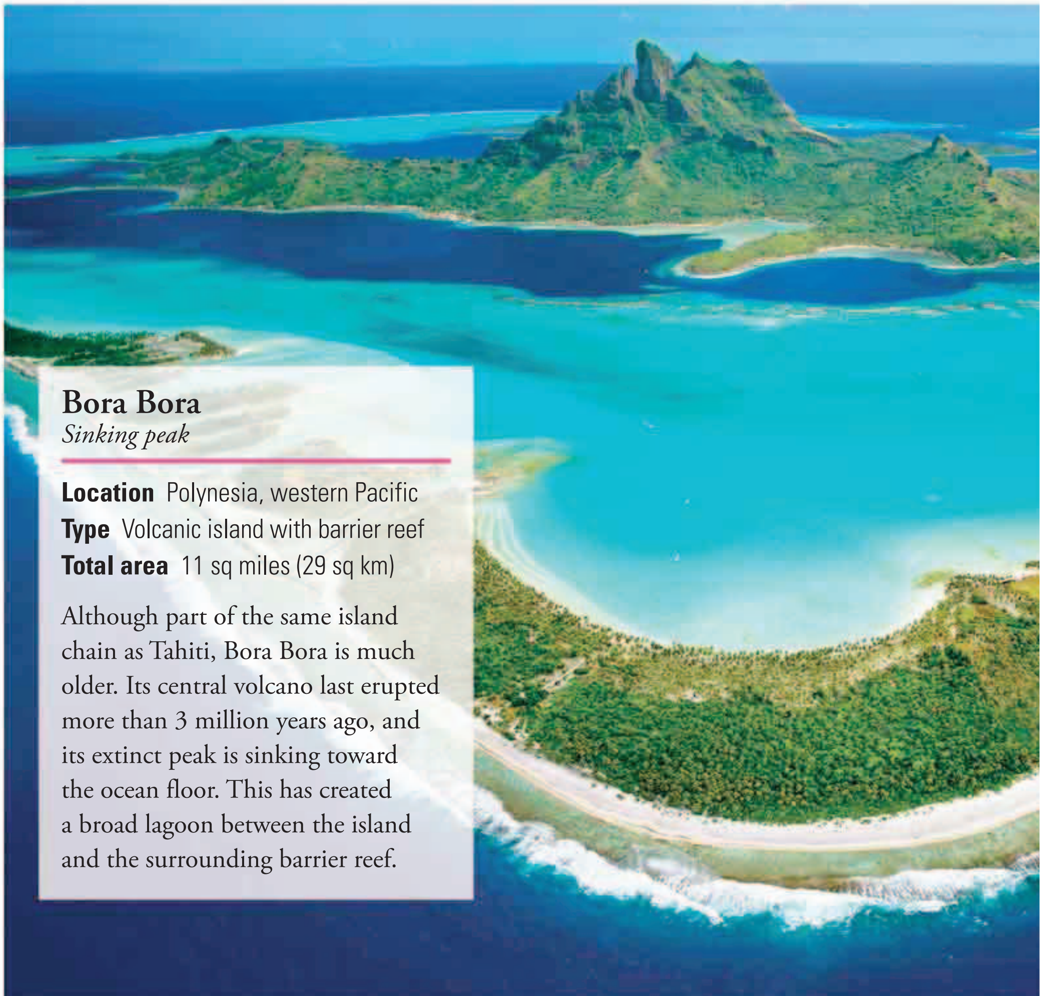


▲ 3. ATOLL
Over millions of years, the volcano disappears, leaving a ring of coral.

Tahiti
High island

Location Polynesia, western Pacific
Type Volcanic island with fringing reef
Total area 403 sq miles (1,045 sq km)

The island of Tahiti is made up of twin volcanic peaks fringed by coral reefs. The volcanoes are extinct and cooling, but still almost as high as when they were active more than 200,000 years ago. As a result, the fringing reefs still lie close to the shore.



Bora Bora
Sinking peak

Location Polynesia, western Pacific
Type Volcanic island with barrier reef
Total area 11 sq miles (29 sq km)

Although part of the same island chain as Tahiti, Bora Bora is much older. Its central volcano last erupted more than 3 million years ago, and its extinct peak is sinking toward the ocean floor. This has created a broad lagoon between the island and the surrounding barrier reef.